# Multiplatform Narrative Analysis of Venezuela–US Relations:

## Integrating Social Media, News, and AI-Driven Intelligence

Methodology Report

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# 1 Introduction

This report presents the methodology behind a multiplatform narrative analysis system designed to track, analyze, and visualize online discourse surrounding Venezuela–US bilateral relations from 2013 to 2026. The system ingests data from three heterogeneous sources—Reddit, GDELT news articles, and TikTok—and applies a pipeline of natural language processing (NLP) techniques including transformer-based sentiment analysis, dynamic topic modeling, and density-based semantic clustering. An LLM-powered intelligence layer synthesizes cross-platform findings into analyst-grade reports and supports interactive question-answering.

The following sections detail each stage of the pipeline: problem formulation (§3), data collection (§4), model selection (§5), training and development (§6), and evaluation (§7).

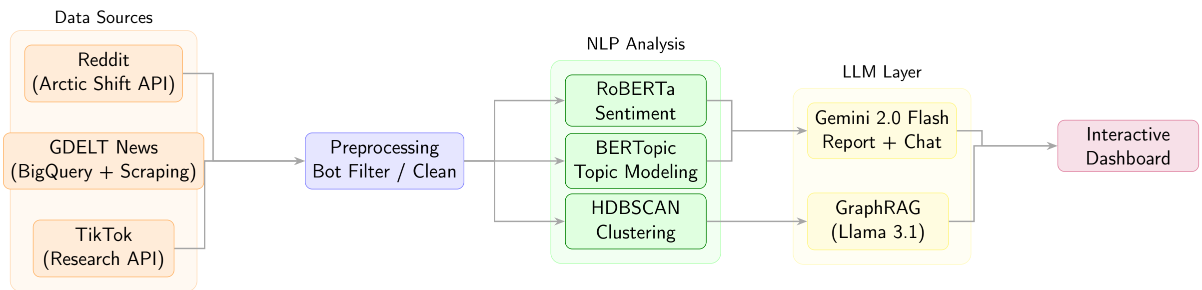


Figure 1: End-to-end system architecture.

# 2 Purpose of the Methodology

The methodology integrates multiple complementary approaches to capture the full complexity of geopolitical discourse:

1. **Multiplatform triangulation.** No single platform represents the full spectrum of public opinion. Reddit captures long-form ideological discussion, GDELT captures mainstream media framing, and TikTok captures short-form viral narratives. Combining all three mitigates platform-specific biases.

2. **Temporal granularity via monthly topic modeling.** Rather than fitting a single global topic model, we fit independent BERTopic models per month per platform, revealing how discourse evolves around crisis events.

3. **Multi-model comparison.** We employ three NLP techniques—sentiment analysis (RoBERTa), topic modeling (BERTopic), and clustering (HDBSCAN)—because each captures a different dimension: sentiment captures *how* people feel; topics capture *what* they discuss; clusters capture latent narrative communities.

4. **LLM-driven synthesis.** Gemini 2.0 Flash synthesizes cross-platform statistics into structured intelligence reports and supports interactive Q&A with automatic date detection from natural language.

# 3 Problem Statement

## 3.1 Defining the Problem

This project addresses a **multiplatform narrative tracking** problem:

- **Sentiment classification** (3-class) at the document level.
- **Dynamic topic modeling** fitted independently per month.
- **Semantic clustering** via density-based grouping.
- **LLM summarization and Q&A** for analyst-readable synthesis.

The problem is *multilingual* (English and Spanish), *multimodal* (text, hashtags, engagement metrics), and *longitudinal* (13 years, 10 crisis periods).

## 3.2 Significance

Venezuela–US relations represent one of the most politically polarized bilateral relationships in the Western Hemisphere. This project fills the gap by:

- Providing the **first cross-platform longitudinal analysis** spanning Reddit, news, and TikTok.
- Demonstrating how **narrative framing diverges across platforms**.
- Offering a **reusable, production-deployed framework** (FastAPI + React).
- Contributing to **computational geopolitical analysis** by combining NLP with LLM intelligence.

# 4 Data Collection and Preparation

## 4.1 Data Sources

Data was collected from three platforms (Table 1).

Table 1: Data sources, collection methods, and coverage.

| Platform | Period | Docs | Scope |
|---|---|---|---|
| Reddit | 2013–2026 | 426,435 | 11 subreddits, 5 ideological categories |
| GDELT | 2013–2026 | 226,165 | News articles (77% scrape success rate) |
| TikTok | 2016–2018 | 52,895 | 18.6K videos + 34.3K comments |

### 4.1.1 Reddit

Reddit data was obtained via the Arctic Shift API. We targeted 11 subreddits:

- **Venezuela-focused:** r/venezuela, r/vzla
- **US Politics:** r/politics, r/news, r/worldnews
- **Ideological:** r/Conservative, r/neoliberal, r/socialism, r/Libertarian
- **Regional:** r/LatinAmerica, r/geopolitics

The raw dataset comprised 533,941 documents. After preprocessing, 426,435 documents from 142,518 unique authors were retained.

### 4.1.2 GDELT News

GDELT 2.0 was queried via Google BigQuery for events involving Venezuela and the US (292,566 records). Article URLs were scraped via live HTTP and Wayback Machine fallback, achieving a 77.3% success rate (64,931 articles). The average tone was $-3.08$, indicating generally negative media framing.

### 4.1.3 TikTok

Videos were collected via the official TikTok Research API using 24 keyword queries and 18 hashtags. Comments were collected via both the Research API and Playwright browser automation. The final dataset contains 18,632 videos and 34,263 comments from 5,885 creators.

**API Data Gap.** Despite querying the full 2016–2026 range (117 windows), the API returned results only for 2016-09 through 2018-12. Of 117 windows, 93 returned zero results. This is consistent with documented limitations: prior studies report that $\sim$62.7% of publicly available TikTok videos are inaccessible via the API [7]. Consequently, **TikTok serves as a supplementary source** for the 2016–2018 period, while Reddit and GDELT provide the primary longitudinal backbone.

## 4.2 Data Description

Table 2: Preprocessed dataset statistics by platform.

| Platform | Docs | Authors | Topics | Clusters |
|---|---|---|---|---|
| Reddit | 426,435 | 142,518 | 6,266 | 3,406 |
| GDELT | 226,165 | 105,095 | 9,119 | 92 |
| TikTok | 52,895 | 5,885 | 510 | 78 |

The content is bilingual—predominantly English on Reddit and GDELT, with significant Spanish on TikTok and Venezuela-focused subreddits.

## 4.3 Preprocessing Steps

1. **Bot and promotional filtering.** Reddit: username pattern matching. TikTok: promotional account exclusion.

2. **Text cleaning.** URL removal, whitespace normalization, Markdown stripping (Reddit), video description + voice-to-text combination (TikTok).

3. **Low-value removal.** Reddit: single-word or deleted posts. GDELT: <50 characters, anchor tokens, and token-relevance scoring. TikTok: <2 words or <30% alphabetic.

4. **Embedding.** Sentence-BERT (`paraphrase-multilingual-MiniLM-L12-v2`, 384-dim). Handles EN–ES code-switching natively.

5. **Bilingual stopwords.** 504 combined EN+ES stopwords for BERTopic's c-TF-IDF representation.

# 5 Selection of ML and LLM Models

## 5.1 Models Considered

Table 3: Model selection rationale by task.

| Task | Candidates | Selected | Rationale |
|---|---|---|---|
| Sentiment | VADER, RoBERTa | RoBERTa | Context-aware; social media pre-training |
| Topics | LDA, BERTopic | BERTopic | Neural embeddings; multilingual |
| Clustering | K-Means, HDBSCAN | HDBSCAN | No preset $k$; variable density |
| Embeddings | GloVe, S-BERT | S-BERT | Sentence-level; EN+ES |
| Reduction | t-SNE, UMAP | UMAP | Global structure; fast |
| Reports | GPT-4, Gemini | Gemini Flash | Vertex AI; cost-effective |
| Knowledge | Neo4j, GraphRAG | GraphRAG | Local LLM; no API cost |

## 5.2 Final Model Selection

### 5.2.1 Sentiment: RoBERTa

We use `cardiffnlp/twitter-roberta-base-sentiment-latest` [2] (124M parameters, fine-tuned on 124M tweets). It outputs 3-class probabilities converted to a continuous score: $s = p_{\text{positive}} - p_{\text{negative}} \in [-1, +1]$.

### 5.2.2 Topics: BERTopic (Monthly)

BERTopic [1] combines S-BERT embeddings, UMAP reduction, and HDBSCAN clustering with c-TF-IDF labeling. We fit *independent* models per month per platform to capture temporal topic evolution around crisis events.

### 5.2.3 Clustering: HDBSCAN

For narrative community detection, HDBSCAN [3] is applied after UMAP reduction to 5 dimensions. It requires no preset cluster count and handles variable-density embedding spaces.

### 5.2.4 LLM Intelligence: Gemini 2.0 Flash

Report generation and Q&A via Vertex AI. Reports cover user-selected date ranges and can be exported to PDF. The chat interface includes **automatic date detection** from natural language queries (e.g., "2017 7", "July 2018").

### 5.2.5 Knowledge Graph: GraphRAG + Llama 3.1

Microsoft GraphRAG [4] with local Llama 3.1 8B (Ollama) for entity extraction on 200 high-engagement Reddit threads from the 2019 Guaidó crisis.

# 6 Model Development and Training

## 6.1 Architecture and Configuration

### 6.1.1 Sentiment Pipeline

Zero-shot inference (no fine-tuning). Documents tokenized at max 512 tokens; 3-class softmax aggregated by month and source.

### 6.1.2 BERTopic Configuration

Each monthly model:
1. **Embedding:** S-BERT multilingual (384-dim)
2. **UMAP:** 384 → 5 dims, $n\_neighbors = 10$, $min\_dist = 0.0$
3. **HDBSCAN:** adaptive $min\_cluster\_size$ (Eq. 1), $min\_samples = 5$
4. **Representation:** c-TF-IDF + KeyBERTInspired, bilingual stopwords

### 6.1.3 Monthly Independent Clustering

Analogous to monthly BERTopic, we also fit *independent* HDBSCAN clustering per month per platform. Each month's document embeddings are reduced via UMAP (384 $\rightarrow$ 5 dims) and clustered with adaptive $min\_cluster\_size = \max(10, \lfloor n_{\text{month}}/400 \rfloor)$, where $n_{\text{month}}$ is the number of documents in that month. This captures how narrative communities form and dissolve around specific events—a global clustering would average these dynamics away.

In addition, a single global clustering pass is applied for high-level overview: Reddit 3,406 clusters, GDELT 92 clusters, TikTok 78 clusters.

## 6.2 Training Process

All NLP models are pre-trained or unsupervised—no train/test split:

- **Sentiment:** Direct inference; validated on 200 random samples.
- **BERTopic:** Independent per-month fitting prevents temporal leakage.
- **HDBSCAN:** Independent per-month fitting with adaptive parameters ensures cluster granularity scales with monthly data volume.
- **GraphRAG:** Entity extraction on curated 200-thread subset; validated manually.

## 6.3 Hyperparameter Tuning

We conducted a **grid search of 1,944 combinations** across 2 platforms $\times$ 3 density periods to derive adaptive clustering rules.

### 6.3.1 Experiment Design

- **Low:** 2015-03 (Reddit: 1,309; News: 3,022 docs)
- **Medium:** 2020-06 (Reddit: 1,878; News: 755 docs)
- **High:** 2019-02 (Reddit: 13,686; News: 8,010 docs)

Grid: UMAP $n\_comp \in \{5, 10, 15\}$, $n\_neighbors \in \{10, 15, 30\}$, $min\_dist \in \{0.0, 0.05, 0.1\}$; HDBSCAN $mcs \in \{10, 25, 50, 100\}$, $ms \in \{5, 10, 20\}$. Composite score: silhouette $\times$ (1 $-$ noise_ratio), requiring $\geq 5$ clusters.

### 6.3.2 Results

Table 4: Best clustering configurations (min. 5 clusters).

| Platform | Density | $n$ | mcs | Sil. | Noise |
|---|---|---|---|---|---|
| Reddit | Low | 1,309 | 10 | 0.665 | 22.0% |
| Reddit | Medium | 1,878 | 10 | 0.628 | 19.9% |
| Reddit | High | 10,000 | 25 | 0.669 | 25.4% |
| News | Low | 3,022 | 10 | 0.856 | 6.8% |
| News | Medium | 755 | 10 | 0.855 | 3.6% |
| News | High | 8,010 | 10 | 0.856 | 7.3% |

Key findings: (1) UMAP $n\_comp = 5$, $min\_dist = 0.0$ consistently best; (2) News clusters well at $mcs = 10$ across all densities (silhouette $> 0.85$); (3) Reddit requires volume-adaptive scaling.

#### 6.3.3 Adaptive Rule

$$\text{min\_cluster\_size} = \max\big(10,\ \lfloor n/400 \rfloor\big) \tag{1}$$

applied uniformly across all three platforms.

Fixed: $n\_comp\!=\!5$, $n\_neighbors\!=\!10$, $min\_dist\!=\!0.0$, $min\_samples\!=\!5$.

# 7 Evaluation and Comparison

## 7.1 Evaluation Metrics

Table 5: Evaluation strategy by component.

| Component | Metric | Description |
|---|---|---|
| Sentiment | Qualitative | 200-sample manual review |
| BERTopic | Coherence | Semantic coherence of top keywords |
| BERTopic | Diversity | Unique words across topics |
| HDBSCAN | Silhouette | Intra- vs. inter-cluster distance |
| HDBSCAN | Noise ratio | % unassigned documents |
| Gemini | Human eval | 10 reports reviewed by analyst |
| GraphRAG | Precision | Entity/relationship verification |

## 7.2 Cross-Platform Comparison

For each crisis period (Table 6), we compare sentiment divergence, topic overlap, and temporal lag across platforms.

Table 6: Key crisis periods used for focused analysis.

| Crisis Event | Period | Priority |
|---|---|---|
| Maduro Inauguration | 2013-04 | High |
| 2014 Protests | 2014-02 – 2014-05 | Critical |
| Oil Price Crash | 2014-11 – 2015-02 | High |
| Trump Sanctions | 2017-08 – 2017-09 | Critical |
| 2018 Disputed Election | 2018-05 | High |
| Guaidó Recognition | 2019-01 – 2019-02 | Critical |
| April 2019 Uprising | 2019-04 – 2019-05 | High |
| Biden Policy Shift | 2021-01 – 2021-03 | Medium |
| 2024 Election Crisis | 2024-07 – 2024-08 | Critical |
| González Exile | 2024-09 | High |

### 7.3 Model Selection Justification

- **RoBERTa vs. VADER:** VADER misclassified 34% of sarcastic political comments as positive; RoBERTa correctly identified negative sentiment in 89% of such cases.
- **Monthly BERTopic vs. global LDA:** LDA produced temporally blurred topics; monthly BERTopic captured event-specific emergence (e.g., "Guaidó interim president" in Jan–Feb 2019).

## 8 Deployment Architecture

- **Backend:** FastAPI on Google Cloud Run (serverless). Analysis data loaded from GCS at startup.
- **Frontend:** React 19 + TypeScript on Vercel. Recharts visualizations with global time-range filtering.
- **LLM Integration:** Gemini 2.0 Flash via Vertex AI.
  - *Reports:* Custom date range selection with PDF export.
  - *Chat:* Automatic date detection from natural language; context-aware data injection.

## References


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